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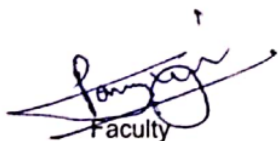
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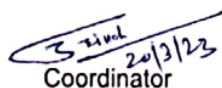
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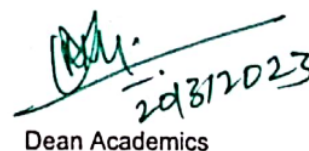
Unit: III

Date of Submission: 20/03/23

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Faculty


Coordinator
20/3/23


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UNIT-3 (Transformers)

LECTURE NO:-1

Operating Principles and Construction

What is a Transformer?

A **transformer** is a static device having two or more electrical circuits separated by a common magnetic core for transferring electrical energy between them at constant frequency.

The use of transformers in transmission system is shown in the Figure below.

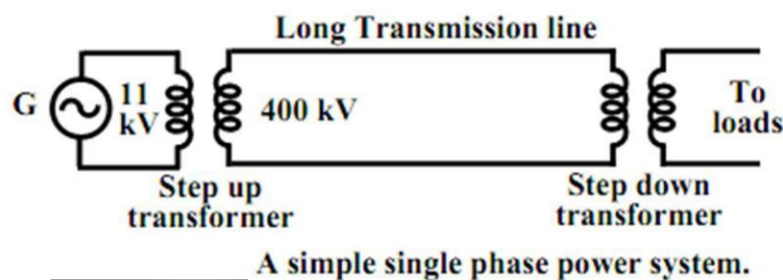


Fig.3-1

Principle of Operation

A transformer in its simplest form will consist of a rectangular laminated magnetic structure on which two coils of different number of turns are wound as shown in Figure 3.2a.

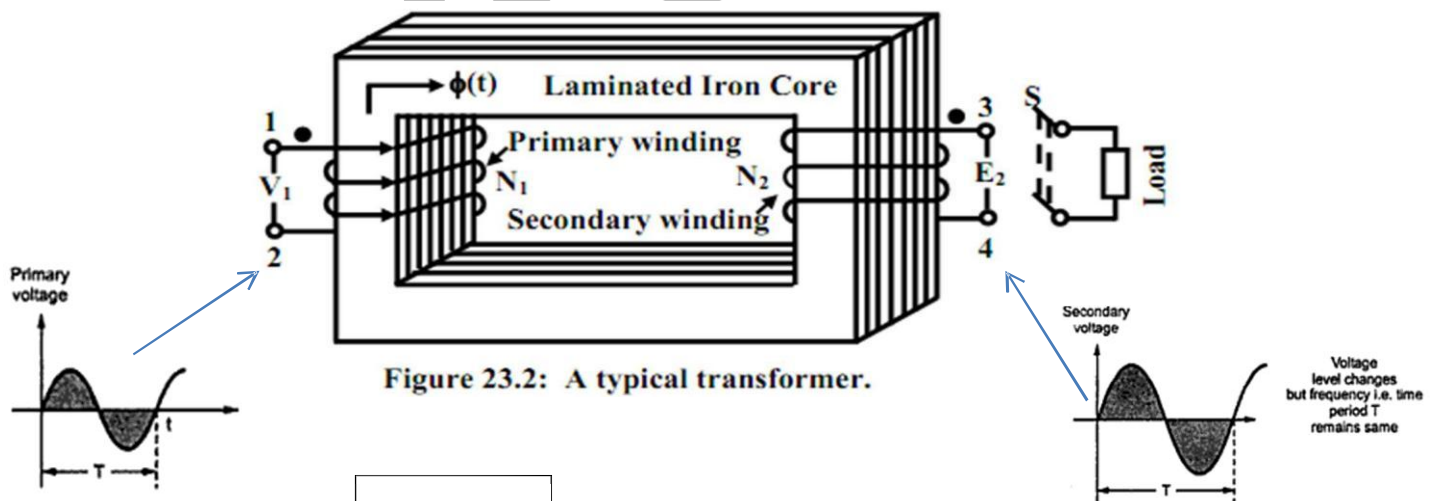


Fig.3-2a

The winding to which AC voltage is impressed is called **the primary** of the transformer and the winding across which the load is connected is called **the secondary** of the transformer.

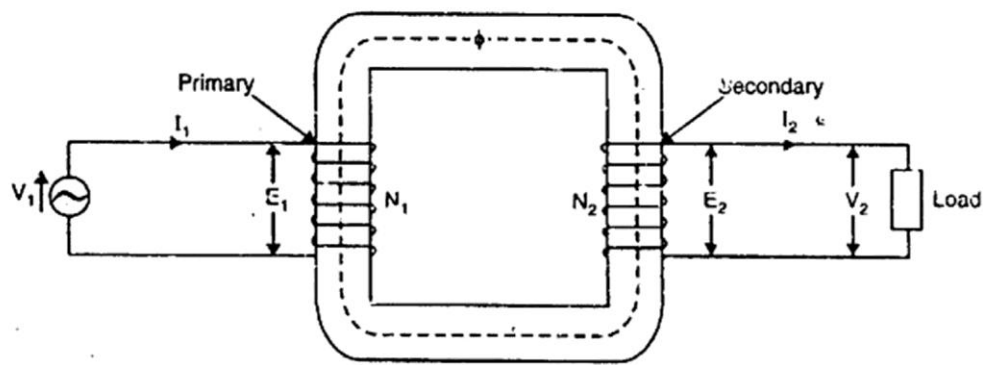


Fig.3-2b

Depending upon the number of turns of the primary (N_1) and secondary (N_2), an alternating emf (E_2) is induced in the secondary. This induced emf (E_2) in the secondary causes a secondary current I_2 . Consequently, terminal voltage V_2 will appear across the load. If $V_2 > V_1$, it is called a **step up-transformer**. On the other hand, if $V_2 < V_1$, it is called a **step-down transformer**.

When an alternating voltage V_1 is applied to the primary, an alternating flux Φ is set up in the core. This alternating flux links both the windings and induces emfs E_1 and E_2 in them according to **Faraday's laws of electromagnetic induction**. The emf E_1 is termed as primary emf and emf E_2 is termed as Secondary emf.

Clearly,
$$E_1 = -N_1 \frac{d\phi}{dt}$$

and
$$E_2 = -N_2 \frac{d\phi}{dt}$$

$$\therefore \frac{E_2}{E_1} = \frac{N_2}{N_1}$$

Note that magnitudes of E_2 and E_1 depend upon the number of turns on the secondary and primary respectively. If $N_2 > N_1$, then $E_2 > E_1$ (or $V_2 > V_1$) and we get a step-up transformer. On the other hand, if $N_2 < N_1$, then $E_2 < E_1$ (or $V_2 < V_1$) and we get a step-down transformer. If load is connected across the secondary winding, the secondary e.m.f. E_2 will cause a current I_2 to flow through the load. Thus, a transformer enables us to transfer a.c. power from one circuit to another with a change in voltage level.

The following points may be noted carefully:

- (i) The transformer action is based on the laws of **electromagnetic induction**.
- (ii) There is no electrical connection between the primary and secondary.
- (iii) There is no change in frequency i.e., output power has the same frequency as the input power.

Can DC Supply be used for Transformers?

The DC supply cannot be used for the transformers. This is because the transformer works on the principle of **mutual induction**, for which current in one coil must change uniformly. If DC supply is given, the current will not change due to constant supply and transformer will not work

There can be saturation of the core due to which transformer draws very large current from the supply when connected to DC.

Thus DC supply should not be connected to the transformers.

Construction

We usually design a power transformer so that it approaches the characteristics of an ideal transformer. To achieve this, following design features are incorporated:

- (i) The core is made of silicon steel which has low hysteresis loss and high permeability. Further, core is laminated in order to reduce eddy current loss. These features considerably reduce the iron losses and the no-load current.
- (ii) Instead of placing primary on one limb and secondary on the other, it is a usual practice to wind one-half of each winding on one limb. This ensures tight coupling between the two windings. Consequently, leakage flux is considerably reduced.

- (iii) The winding resistances are minimized to reduce Copper loss and resulting rise in temperature and to ensure high efficiency.

Transformers are of two types: (i) core-type transformer (see Fig.3-3) and (ii) shell-type transformer (see Fig.3-4).

Core-Type Transformer: In a core-type transformer, half of the primary winding and half of the secondary winding are placed round each limb to reduce the leakage flux.

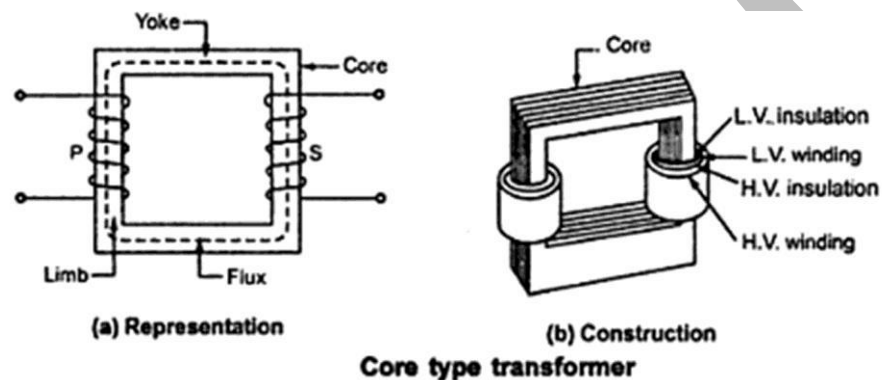


Fig.3-3

Shell-Type Transformer: This method of construction involves the use of a double magnetic circuit. Both the windings are placed round the central limb to ensure a low-reluctance flux path.

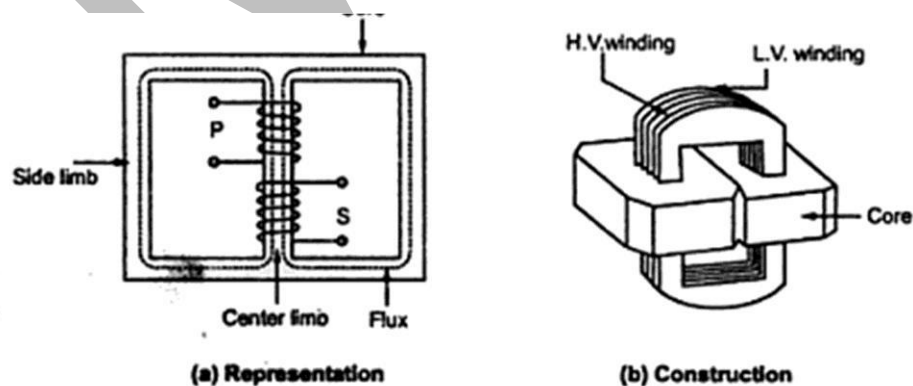


Fig.3-4

LECTURE NO:-2**Comparison of Core and Shell Type Transforms**

Core Type	Shell Type
The winding encircles the core.	The core encircles most part of the winding
It has single magnetic circuit	It has double magnetic circuit
The core has two limbs	The core has three limbs
The cylindrical coils are used.	The multilayer, disc or sandwich type coils are used.
The winding are uniformly distributed on two limbs hence natural cooling is effective	The natural cooling does not exist as the windings are surrounded by the core.
Preferred for low voltage transformers.	Preferred for high voltage transformers.

Cooling of Transformers

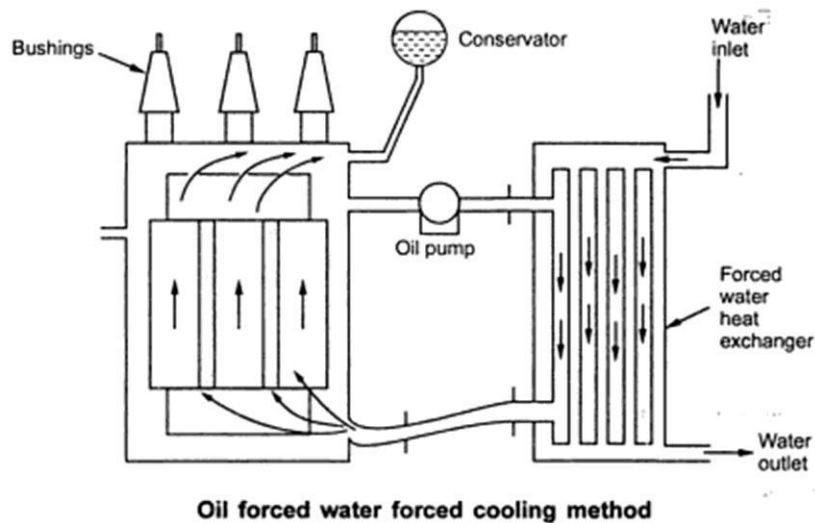
When transformer supplies a load, two types of losses occur inside the transformer. The iron losses occur in the core while copper losses occur in the windings. The power lost due to these losses appears in the form of heat. This heat increases the temperature of the transformer. **To keep the temperature rise of the transformer within limits, a suitable coolant and cooling method is necessary.**

The various cooling methods are designated witch depended upon:

A: cooling medium used and B: type of circulation employed.

The various coolant used such as Air, Gas, Mineral oil, and water.

One of cooling method system is shown in figure below which is called **Oil Forced Water Forced cooling system**;



EMF Equation of a Transformer

Consider that an alternating voltage V_1 of frequency f is applied to the primary as shown in Fig. 3-2b. The sinusoidal flux Φ produced by the primary can be represented as:

$$\phi = \phi_m \sin \omega t$$

The instantaneous e.m.f. e_1 induced in the primary is

$$\begin{aligned} e_1 &= -N_1 \frac{d\phi}{dt} = -N_1 \frac{d}{dt} (\phi_m \sin \omega t) \\ &= -\omega N_1 \phi_m \cos \omega t = -2\pi f N_1 \phi_m \cos \omega t \\ &= 2\pi f N_1 \phi_m \sin(\omega t - 90^\circ) \end{aligned} \quad (i)$$

It is clear from the above equation that maximum value of induced e.m.f. in the primary is

$$E_{m1} = 2\pi f N_1 \phi_m$$

The r.m.s. value E of the primary e.m.f. is

$$E_1 = \frac{E_{m1}}{\sqrt{2}} = \frac{2\pi f N_1 \phi_m}{\sqrt{2}}$$

or $E_1 = 4.44 f N_1 \phi_m$

Similarly $E_2 = 4.44 f N_2 \phi_m$

In an ideal transformer, $E_1 = V_1$ and $E_2 = V_2$.

Note. It is clear from exp. (i) above that e.m.f. E_1 induced in the primary lags behind the flux ϕ by 90° . Likewise, e.m.f. E_2 induced in the secondary lags behind flux ϕ by 90° .

Voltage Transformation Ratio (K)

From the above equations of induced emf, we have,

$$\frac{E_2}{E_1} = \frac{N_2}{N_1} = K$$

The constant K is called voltage transformation ratio. Thus if $K = 5$ (i.e.

$N_2/N_1 = 5$), then $E_2 = 5 E_1$.

Concept of Ideal Transformer

A transformer is said to be ideal if it satisfies following properties:

- i) It has no losses.
- ii) Its windings have zero resistance.
- iii) Leakage flux is zero i.e. 100 % flux produced by primary links with the secondary.
- iv) Permeability of core is so high that negligible current is required to establish the flux in it.

NOTE:

For an ideal transformer, the primary applied voltage V_1 is same as the primary induced emf E_1 as there are no voltage drops.

For ideal transformer:

- (i) $E_1 = V_1$ and $E_2 = V_2$ as there is no voltage drop in the windings.

$$\therefore \frac{E_2}{E_1} = \frac{V_2}{V_1} = \frac{N_2}{N_1} = K$$

- (ii) there are no losses. Therefore, volt-amperes input to the primary are equal to the output volt-amperes i.e.

$$V_1 I_1 = V_2 I_2$$

$$\text{or } \frac{I_2}{I_1} = \frac{V_1}{V_2} = \frac{1}{K}$$

Hence, currents are in the inverse ratio of voltage transformation ratio. This simply means that if we raise the voltage, there is a corresponding decrease of current.

Volt-Ampere Rating

Transformer rating is specified as the product of voltage and current and called **VA rating**.

$$\text{kVA rating of a transformer} = \frac{V_1 I_1}{1000} = \frac{V_2 I_2}{1000}$$

The full load primary and secondary currents which indicates the safe maximum values of currents which transformer windings can carry can be given as:

$I_1 \text{ full load} = \frac{\text{kVA rating} \times 1000}{V_1} \quad \dots \text{ (1000 to convert kVA to VA)}$
$I_2 \text{ full load} = \frac{\text{kVA rating} \times 1000}{V_2}$

Ideal Transformer on No Load

Consider an ideal transformer in Fig. 3-5. For no load $I_2 = 0$. I_1 is just necessary to produce flux in the core, which is called **magnetising** current denoted as I_m . I_m is very small and lags V_1 by 90° as the winding is purely inductive.

According to Lenz's law, the induced e.m.f opposes the cause producing it which is supply voltage V_1 . Hence E_1 and E_2 are in antiphase with V_1 but equal in magnitude and E_1 and E_2 are in phase.

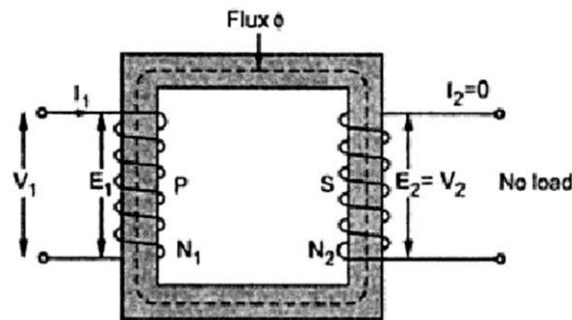
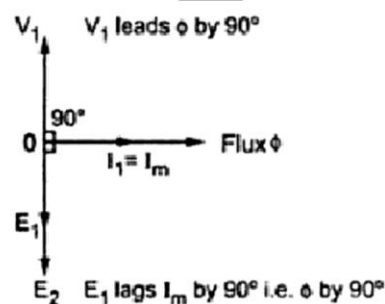


Fig.3-5

This can be illustrated in the phase diagram as shown below:

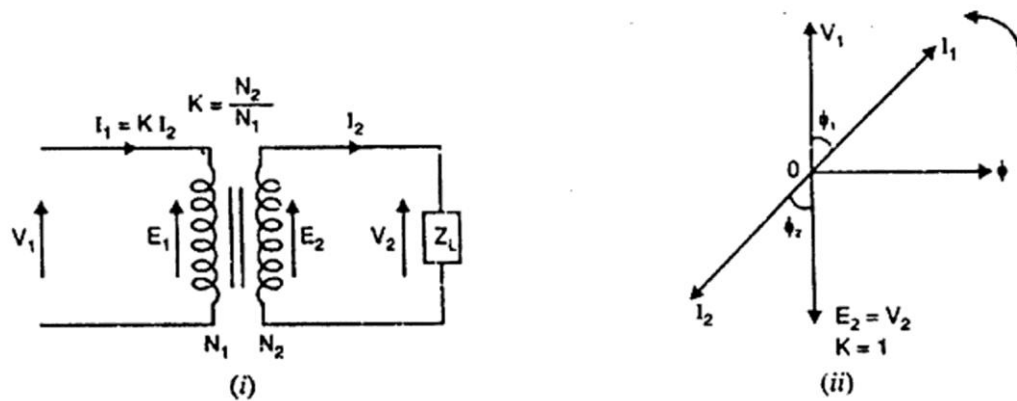


Phasor diagram for ideal transformer on no load

Ideal Transformer on Load

Let us connect a load Z_L across the secondary of an ideal transformer as shown in Figure below:

The secondary emf E_2 will cause a current I_2 to flow through the load:



$$I_2 = \frac{E_2}{Z_L} = \frac{V_2}{Z_L}$$

The angle at which I_2 leads or lags V_2 (or E_2) depends upon the resistance and reactance of the load. In the present case, we have considered inductive load so that current I_2 lags behind V_2 (or E_2) by ϕ_2 .

The secondary current I_2 sets up an m.m.f. $N_2 I_2$ which produces a flux in the opposite direction to the flux ϕ originally set up in the primary by the magnetizing current. This will change the flux in the core from the original value. However, the flux in the core should not change from the original value.

Thus when a transformer is loaded and carries a secondary current I_2 , then a current I_1 , ($= K I_2$) must flow in the primary to maintain the m.m.f. balance. In other words, the primary must draw enough current to neutralize the demagnetizing effect of secondary current so that mutual flux ϕ remains constant. Thus as the secondary current increases, the primary current I_1 ($= K I_2$) increases in unison and keeps the mutual flux ϕ constant. The power input, therefore, automatically increases with the output. For example if $K = 2$ and $I_2 = 2\text{A}$, then primary will draw a current $I_1 = K I_2 = 2 \times 2 = 4\text{A}$. If secondary current is increased to 4A , then primary current will become $I_1 = K I_2 = 2 \times 4 = 8\text{A}$.

The Phasor diagram for the ideal transformer on load is shown in Figure (ii) above.

The secondary current I_2 lags behind V_2 (or E_2) by Φ . It causes a primary current $I_1 = K I_2 = I_2$ (for $K=1$) which is in antiphase with it.

$$(i) \quad \phi_1 = \phi_2$$

$$\text{or} \quad \cos \phi_1 = \cos \phi_2$$

Thus, power factor on the primary side is equal to the power factor on the secondary side.

(ii) Since there are no losses in an ideal transformer, input primary power is equal to the secondary output power i.e.,

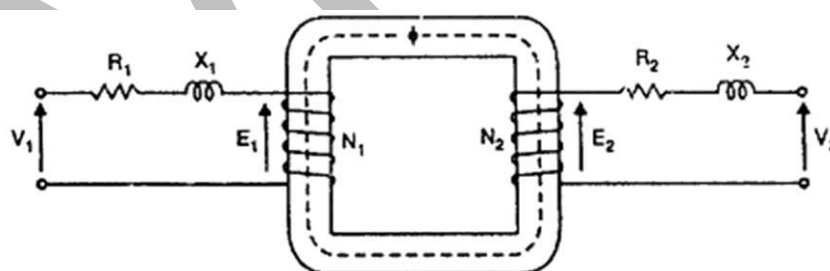
$$V_1 I_1 \cos \phi_1 = V_2 I_2 \cos \phi_2$$

✓ Practical Transformer

A practical transformer differs from the ideal transformer in many respects. The practical transformer has (i) **iron losses** (ii) **winding resistances** and (iii) **magnetic leakage**, giving rise to leakage reactance.

(i) **Iron losses.** Since the iron core is subjected to alternating flux, there occurs eddy current and hysteresis loss in it.

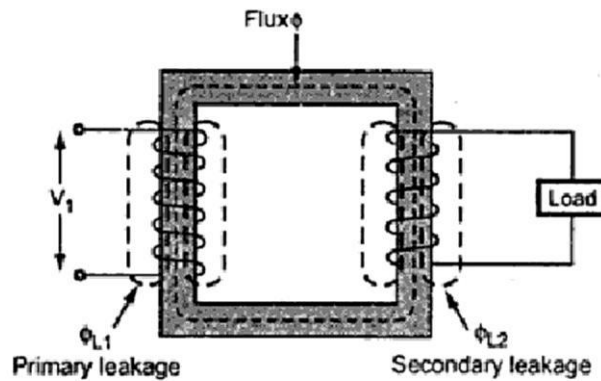
(ii) **Winding resistances.** Since the windings consist of copper conductors, it immediately follows that both primary and secondary will have winding resistance. The primary resistance R_1 and secondary resistance R_2 act in series with the respective windings as shown below:



(iii) **Leakage reactance.** Both primary and secondary currents produce flux. The flux Φ which links both the windings is the useful flux. However, primary current would produce some flux Φ which would not link the secondary winding and is called **mutual flux** (for more information review Lecture Note 2) (See Fig. below).

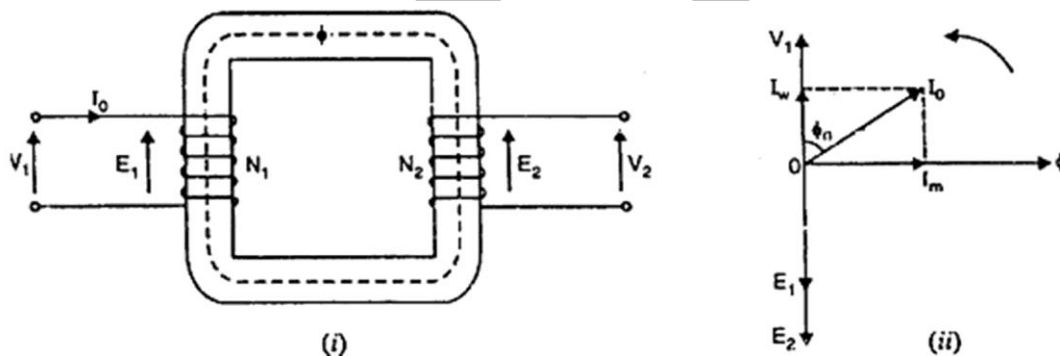
LECTURE NO:-3

Practical Transformer on No Load



Practical Transformer on No Load

Consider the figure below:



The primary will draw a small current I_0 to supply (i) the iron losses and (ii) a very small amount of copper loss in the primary. Hence the primary no load current I_0 is not 90° behind the applied voltage V_1 but lags it by an angle $\Phi_0 < 90^\circ$ as shown in the phasor diagram.

The no-load primary current I_0 can be resolved into two rectangular components:

- (i) The component I_w in phase with the applied voltage V_1 . This is known as active or working or iron loss component and supplies the iron loss and a very small primary copper loss.

$$I_w = I_0 \cos \phi_0$$

The component I_m lagging behind V_1 by 90° and is known as magnetizing component. It is this component which produces the mutual flux ϕ in the core.

$$I_m = I_0 \sin \phi_0$$

Clearly, I_0 is phasor sum of I_m and I_w ,

$$\therefore I_0 = \sqrt{I_m^2 + I_w^2}$$

No load p.f., $\cos \phi_0 = \frac{I_w}{I_0}$

It is emphasized here that no load primary copper loss (i.e. $I_0^2 R_1$) is very small and may be neglected. Therefore, the no load primary input power is practically equal to the iron loss in the transformer i.e.,

$$\text{No load input power, } W_0 = \text{Iron loss}$$

Note. At no load, there is no current in the secondary so that $V_2 = E_2$. On the primary side, the drops in R_1 and X_1 , due to I_0 are also very small because of the smallness of I_0 . Hence, we can say that at no load, $V_1 = E_1$.

Practical Transformer on Load

We shall consider two cases (i) when such a transformer is assumed to have no winding resistance and leakage flux (ii) when the transformer has winding resistance and leakage flux.

No winding resistance and leakage flux

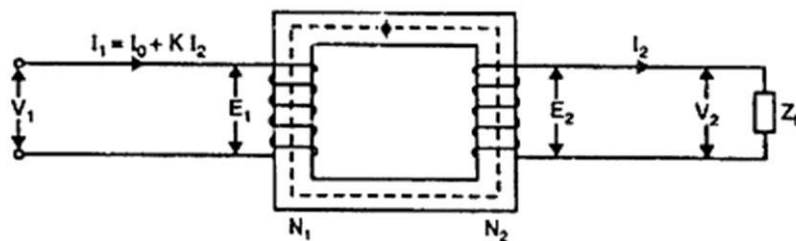


Fig. above shows a practical transformer with the assumption that resistances and leakage reactances of the windings are negligible. With this assumption, $V_2 = E_2$ and $V_1 = E_1$.

Let us take the usual case of inductive load which causes the I_2 to lag V_2 by Φ_2 .

The total primary current I_1 must meet two requirements:

- It must supply the no-load current I_0 to meet the iron losses in the transformer and to provide flux in the core.
- It must supply a current I'_2 to counteract the demagnetizing effect of secondary current I_2 . The magnitude of I'_2 will be such that:

$$N_1 I'_2 = N_2 I_2$$

or
$$I'_2 = \frac{N_2}{N_1} I_2 = K I_2$$

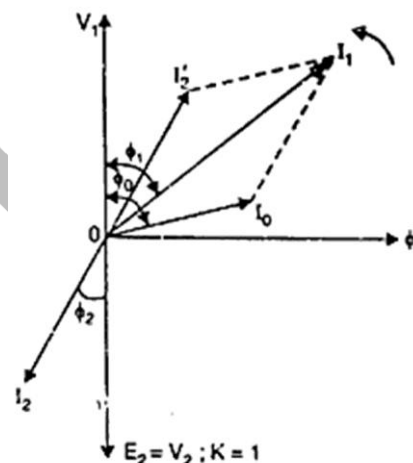
The total primary current I_1 is the phasor sum of I'_2 and I_0 i.e.,

$$I_1 = I'_2 + I_0$$

where $I'_2 = -K I_2$

Note that I'_2 is 180° out of phase with I_2 .

Phasor Diagram: Both E_1 and E_2 lag behind the mutual flux ϕ by 90° . The current I' represents the primary current to neutralize the demagnetizing effect of secondary current I_2 . Now $I'_2 = K I_2$ and is antiphase with I_2 . I_0 is the no-load current of the transformer. The phasor sum of I'_2 and I_0 gives the total primary current I_1 . Note that in drawing the phasor diagram, the value of K is assumed to be unity so that primary phasors are equal to secondary phasors.



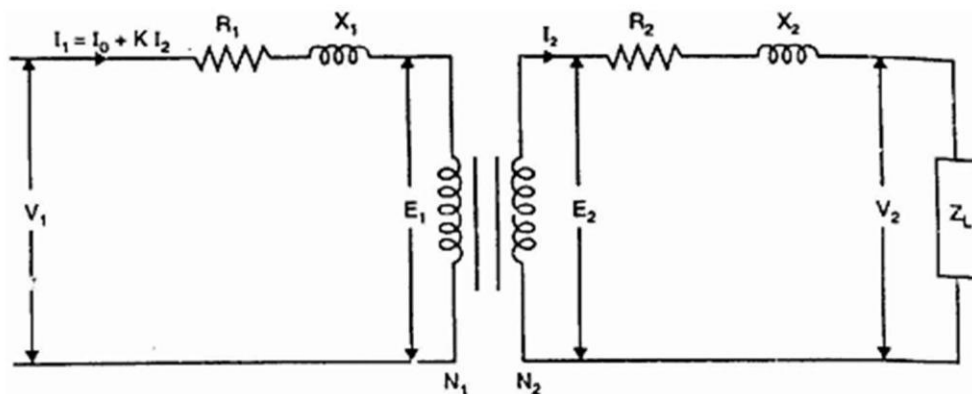
Transformer with resistance and leakage reactance

The total primary current I_1 must meet **two requirements**:

- It must supply the no-load current I_0 to meet the iron losses in the transformer and to provide flux in the core.
- It must supply a current I'_2 to counteract the demagnetizing effect of secondary current I_2 . The magnitude of I'_2 will be such that:

$$N_1 I'_2 = N_2 I_2$$

or
$$I'_2 = \frac{N_2}{N_1} I_2 = K I_2$$



The total primary current I_1 will be the phasor sum of I'_2 and I_0 i.e.,

$$I_1 = I'_2 + I_0 \quad \text{where} \quad I'_2 = -K I_2$$

$$V_1 = -E_1 + I_1(R_1 + jX_1) \quad \text{where} \quad I_1 = I_0 + (-K I_2)$$

$$= -E_1 + I_1 Z_1$$

$$V_2 = E_2 - I_2(R_2 + jX_2)$$

$$= E_2 - I_2 Z_2$$

Phasor Diagram:

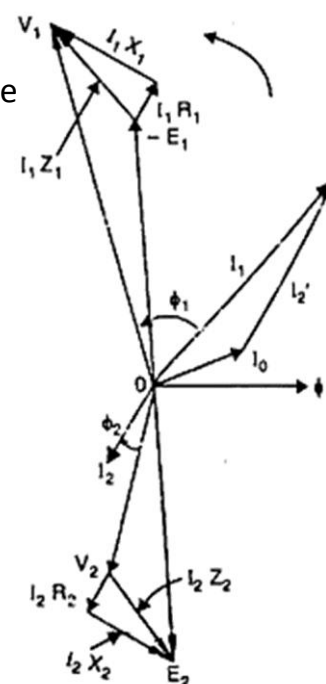
Note that counter emf that opposes the applied voltage V_1 is $-E_1$. Therefore, if we add $I_1 R_1$ (in phase with I_1) and $I_1 X_1$ (90° ahead of I_1) to $-E_1$, we get the applied primary voltage V_1 . The phasor E_2 represents the induced emf in the secondary by the mutual flux. The secondary terminal voltage V_2 will be what is left over after subtracting $I_2 R_2$ and $I_2 X_2$ from E_2 .

$$\text{Load power factor} = \cos \phi_2$$

$$\text{Primary power factor} = \cos \phi_1$$

$$\text{Input power to transformer, } P_1 = V_1 I_1 \cos \phi_1$$

$$\text{Output power of transformer, } P_2 = V_2 I_2 \cos \phi_2$$



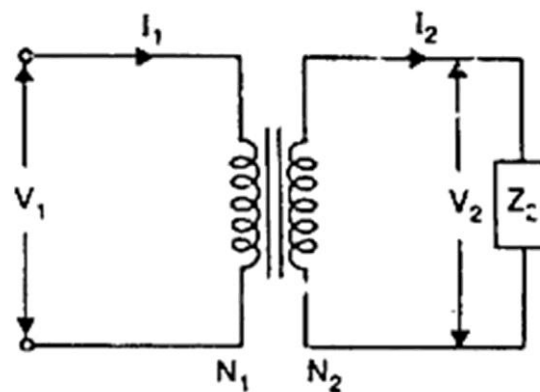
LECTURE NO:-4

Modelling and Equivalent Circuits of Single Phase Transformers

The term equivalent circuit of a transformer means the combination of fixed and variable resistances and reactances, which exactly simulates performance and working of the transformer.

Impedance Ratio

Consider a transformer having impedance Z_2 in the secondary as shown in the figure below:



$$Z_2 = \frac{V_2}{I_2}$$

$$Z_1 = \frac{V_1}{I_1}$$

$$\therefore \frac{Z_2}{Z_1} = \left(\frac{V_2}{V_1} \right) \times \left(\frac{I_1}{I_2} \right)$$

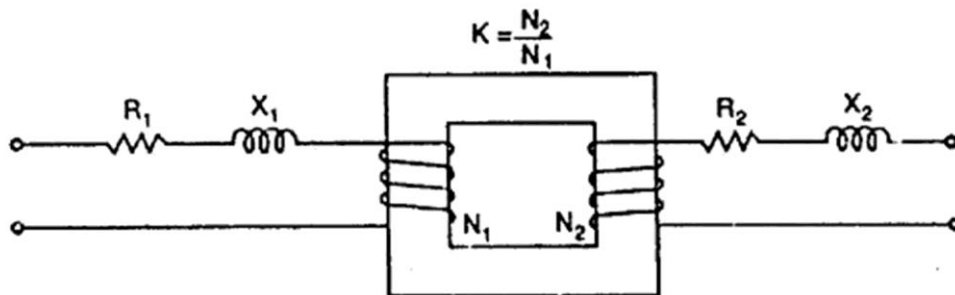
or

$$\boxed{\frac{Z_2}{Z_1} = K^2}$$

Shifting Impedances in a Transformer

NOTE:

Consider the following figure:



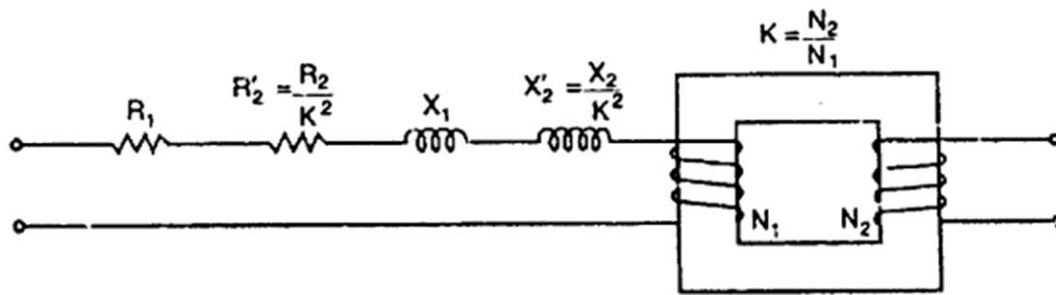
We can transfer the parameters from one winding to the other. Thus:

- ▶ A resistance R_1 in the primary becomes $K^2 R_1$ **when transferred to the secondary.**
- ▶ A resistance R_2 in the secondary becomes R_2/K^2 **when transferred to the primary.**
- ▶ A reactance X_1 in the primary becomes $K^2 X_1$ **when transferred to the secondary.**
- ▶ A reactance X_2 in the secondary becomes X_2/K^2 **when transferred to the primary.**

NOTE:

- ▶ When transferring resistance or reactance from primary to secondary, **multiply it by K^2**
- ▶ When transferring resistance or reactance from secondary to primary, **divide it by K^2**
- ▶ When transferring voltage or current from one winding to the other, **only K is used.**

A- Referred to primary



- Equivalent resistance of transformer referred to primary

$$R_{01} = R_1 + R'_2 = R_1 + \frac{R_2}{K^2}$$

- Equivalent reactance of transformer referred to primary

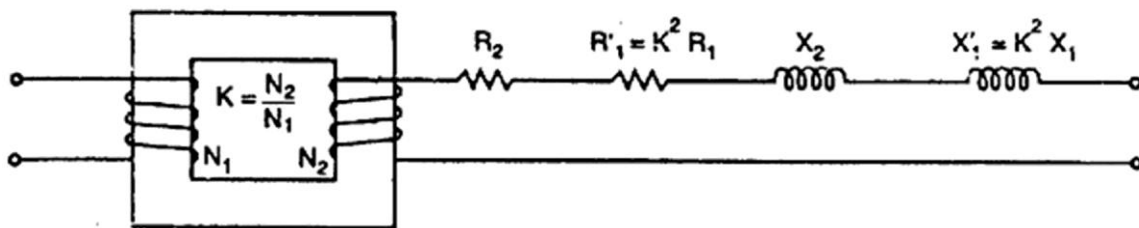
$$X_{01} = X_1 + X'_2 = X_1 + \frac{X_2}{K^2}$$

- Equivalent impedance of transformer referred to primary

$$Z_{01} = \sqrt{R_{01}^2 + X_{01}^2}$$

- The value of primary current \$I_1\$

$$I_1 = KI_2$$

B- Referred to secondary

- Equivalent resistance of transformer referred to secondary

$$R_{02} = R_2 + R'_1 = R_2 + K^2 R_1$$

- Equivalent reactance of transformer referred to secondary

$$X_{02} = X_2 + X'_1 = X_2 + K^2 X_1$$

- Equivalent impedance of transformer referred to secondary

$$Z_{02} = \sqrt{R_{02}^2 + X_{02}^2}$$

- The value of secondary voltage referred to primary

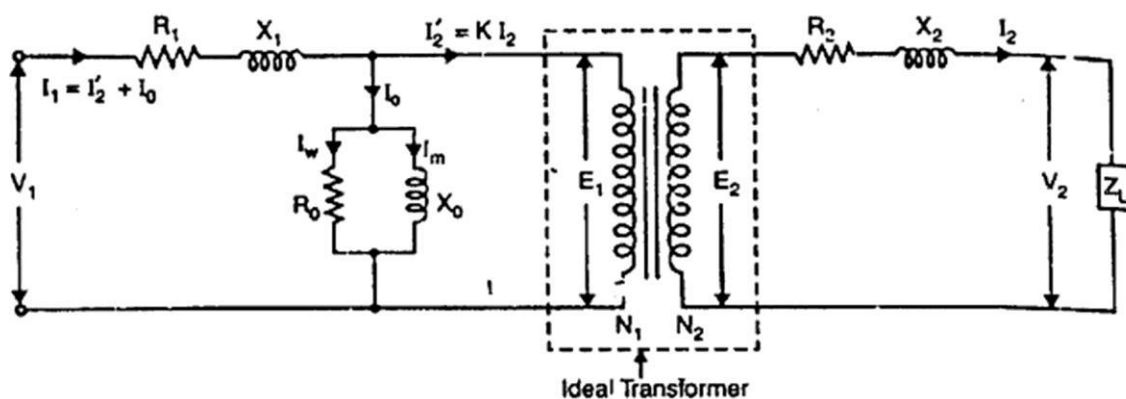
$$V_2 = KV_1$$

What the Importance of Shifting Impedances?

If we shift all the impedances from one winding to the other, the transformer is eliminated and we get an equivalent electrical circuit. ***Various voltages and currents can be readily obtained by solving this electrical circuit.***

Exact Equivalent Circuit of a Loaded Transformer

The equivalent circuit for the transformer can be represented as shown in the figure.



Where:

R_1 : primary winding resistance

R_2 : secondary winding resistance

X_1 : leakage reactance of primary winding

X_2 : leakage reactance of the secondary winding

R_0 : represents the core losses (hysteresis and eddy current losses)

X_0 : represents magnetising reactance of the core

I_m : magnetizing current (to create magnetic flux in the core)

I_w : active current (required to supply the core losses)

I_0 = no load primary current

- **NOTE¹:** Parallel circuit $R_0 - X_0$ is the no-load equivalent circuit of the transformer or called exciting circuit.
- **NOTE²:** The equivalent circuit has created two normal electrical circuits separated only by an ideal transformer whose function is to change values according to the equation:

$$\frac{E_2}{E_1} = \frac{N_2}{N_1} = \frac{I'_2}{I_2}$$

- **NOTE³:** If Z_L is the external load across the secondary circuit, voltage E_2 induced in the secondary by mutual flux will produce a secondary current I_2 , hence:

$$V_2 = E_2 - I_2(R_2 + j X_2) = E_2 - I_2 Z_2$$

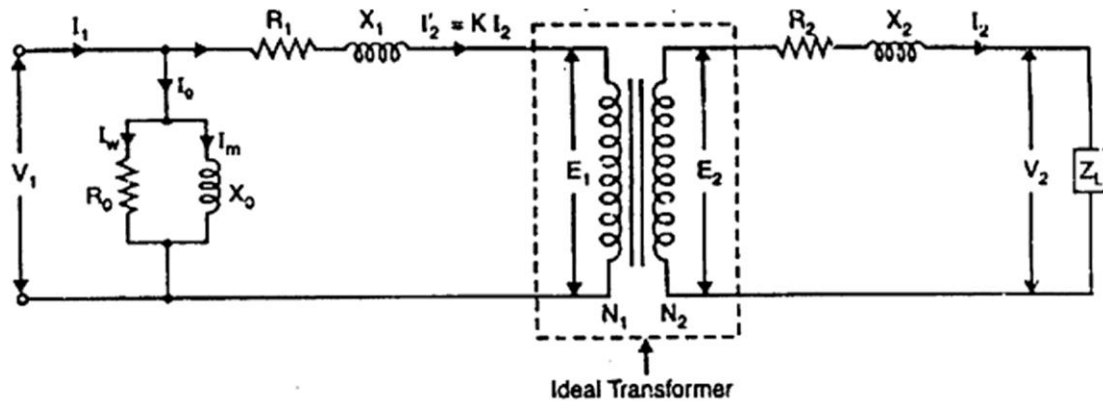
Similarly supply voltage can be given as

$$V_1 = -E_1 + I_1(R_1 + j X_1) = -E_1 + I_1 Z_1$$

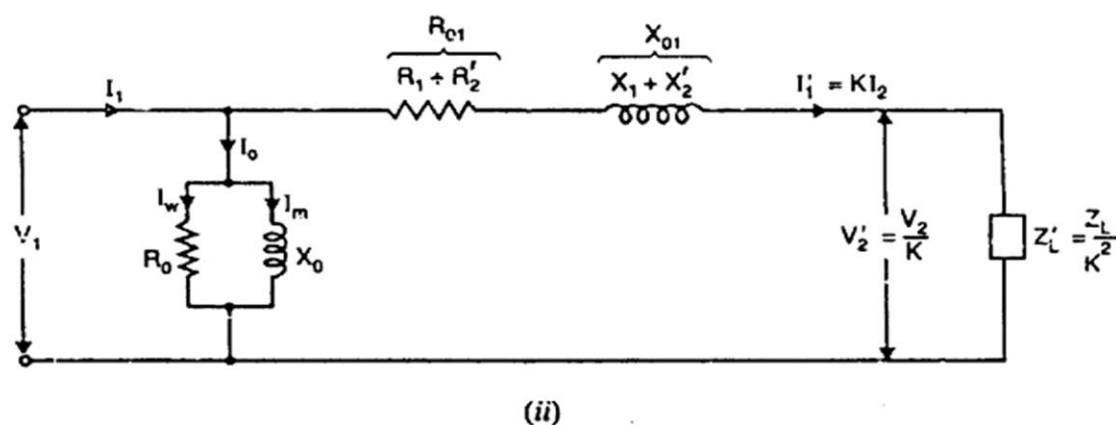
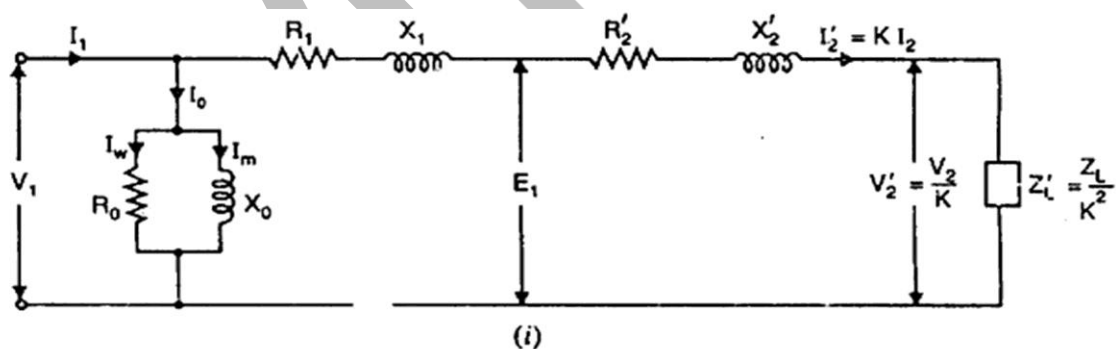
- **NOTE⁵:** When the transformer is loaded to carry the secondary current I_2 , the primary current consists of two components:
 - I_0 to provide magnetizing current and the current required to supply the core losses.
 - primary current I'_2 ($= K I_2$) required to supply the load connected to the secondary

Simplified Equivalent Circuit of a Loaded Transformer

If I_0 of a transformer is small as compared to the rated primary current I_1 , voltage drops in R_1 and X_1 due to I_0 are negligible. Hence, the exact equivalent circuit can be simplified by transferring the shunt circuit $R_0 - X_0$ to the input terminals as shown below:



If all the secondary quantities are referred to the primary, we can get the *simplified equivalent circuit of the transformer referred to the primary* as shown below.



From the above circuits:

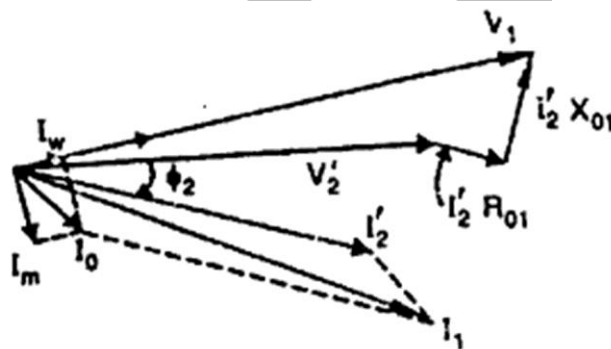
$$R'_2 = \frac{R_2}{K^2}; \quad X'_2 = \frac{X_2}{K^2}; \quad Z'_L = \frac{Z_L}{K^2}; \quad V'_2 = \frac{V_2}{K}; \quad I'_2 = K I_2$$

$$Z_{01} = R_{01} + j X_{01}$$

where

$$R_{01} = R_1 + R'_2; \quad X_{01} = X_1 + X'_2$$

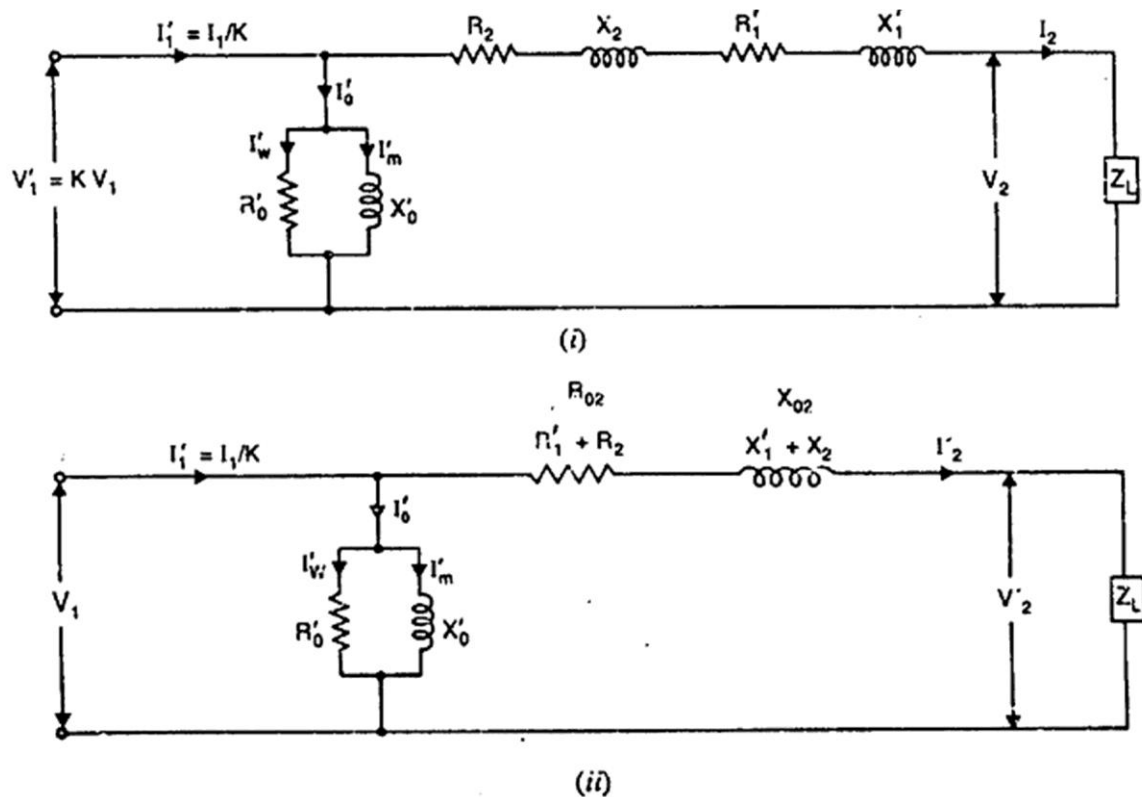
Hence the phasor diagram can be obtained as:



Based on the above phasor diagram we can notice the following:

- ▶ The referred value of load voltage V'_2 is chosen as the reference phasor.
- ▶ I'_2 is lagging V'_2 by phase angle ϕ_2 .
- ▶ $I'_2 R_{01}$ is in phase with I'_2 and the voltage drop $I'_2 X_{01}$ leads I'_2 by 90° .
- ▶ I_w is in phase with V_1 while I_m lags behind V_1 by 90° .
- ▶ The phasor sum of I_w and I_m is I_0 .
- ▶ The phasor sum of I_0 and I'_2 is the input current I_1 .

If all the primary quantities are referred to the secondary, we can get the *simplified equivalent circuit of the transformer referred to the secondary* as shown below



From the above circuit:

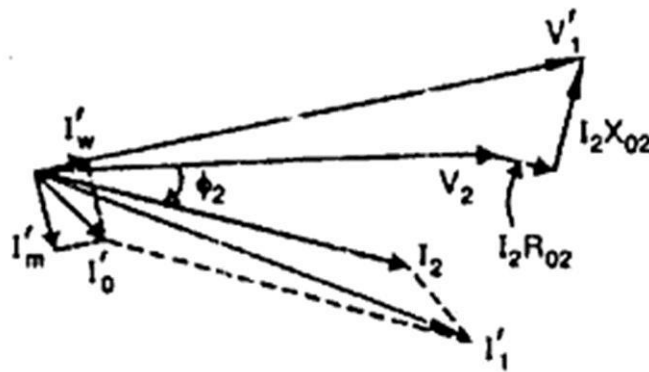
$$R_1' = K^2 R_1; \quad X_1' = K^2 X_1; \quad V_2' = K V_1; \quad I_1' = \frac{I_1}{K}$$

$$Z_{02} = R_{02} + j X_{02}$$

Where

$$R_{02} = R_2 + R_1'; \quad X_{02} = X_2 + X_1'$$

Hence the phasor diagram can be obtained as:



Based on the above phasor diagram we can notice the following:

- ▶ The referred value of load voltage V_2 is chosen as the reference phasor.
- ▶ I_2 is lagging V_2 by phase angle ϕ_2 .
- ▶ $I_2 R_{02}$ is in phase with I_2 and the voltage drop $I_2 X_{02}$, leads I_2 by 90° .
- ▶ I'_w is in phase with V'_1 while I'_m lags behind V'_1 by 90° .
- ▶ The phasor sum of I'_w and I'_m is I'_0 .
- ▶ The phasor sum of I'_0 and I_2 is the input current I'_1 .

LECTURE-5

INTRODUCTION TO DC MACHINES

The operation of the three major classes of electric machines—DC, synchronous, and induction—first is described as intuitively as possible. The second part of the chapter is devoted to a discussion of the applications and selection criteria for the different classes of machines. The emphasis of this chapter is on explaining the properties of each type of machine, with its advantages and disadvantages with regard to other types; and on classifying these machines in terms of their performance characteristics and preferred field of application.

ROTATING ELECTRIC MACHINES

The range of sizes and power ratings and the different physical features of rotating machines are such that the task of explaining the operation of rotating machines in a single chapter may appear formidable at first. Some features of rotating machines, however, are common to all such devices. This introductory section is aimed at explaining the common properties of all rotating electric machines. We begin our discussion with reference to Figure 14.1, in which a hypothetical rotating machine is depicted in a cross-sectional view. In the figure, a box with a cross inscribed in it indicates current flowing into the page, while a dot represents current out of the plane of the page.

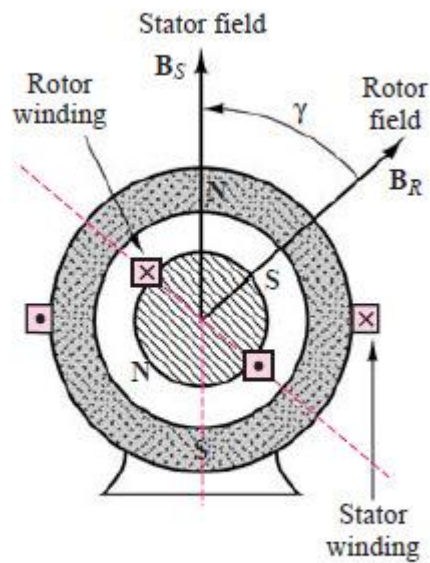


Figure 14.1 A rotating electric machine

In Figure 14.1, we identify a **stator**, of cylindrical shape, and a **rotor**, which, as the name indicates, rotates inside the stator, separated from the latter by means of an air gap. The rotor and stator each consist of a magnetic core, some electrical insulation, and the windings necessary to establish a magnetic flux (unless this is created by a permanent magnet). The rotor is mounted on a bearing-supported shaft, which can be connected to *mechanical loads* (if the machine is a motor) or to a *prime mover* (if the machine is a generator) by means of belts, pulleys, chains, or other mechanical couplings. The windings carry the electric currents that generate the magnetic fields and flow to the electrical loads, and also provide the closed loops in which voltages will be induced.

BASIC CLASSIFICATION OF ELECTRIC MACHINES

An immediate distinction can be made between different types of windings characterized by the nature of the current they carry. If the current serves the sole

purpose of providing a magnetic field and is independent of the load, it is called a *magnetizing*, or excitation, current, and the winding is termed a **field winding**. Field currents are nearly always direct current (DC) and are of relatively low power, since their only purpose is to magnetize the core (recall the important role of high-permeability cores in generating large magnetic fluxes from relatively small currents). On the other hand, if the winding carries only the load current, it is called an **armature**. In DC and alternating-current (AC) synchronous machines, separate windings exist to carry field and armature currents. In the induction motor, the magnetizing and load currents flow in the same winding, called the *input winding*, or *primary*; the output winding is then called the *secondary*. As we shall see, this terminology, which is reminiscent of transformers, is particularly appropriate for induction motors.

It is also useful to classify electric machines in terms of their energy conversion characteristics. A machine acts as a **generator** if it converts mechanical energy from a prime mover, say, an internal combustion engine, to electrical form. Examples of generators are the large machines used in power generating plants, or the common automotive alternator. A machine is classified as a **motor** if it converts electrical energy to mechanical form. The latter class of machines is probably of more direct interest to you, because of its widespread application in engineering practice. Electric motors are used to provide forces and torques to generate motion in countless industrial applications. Machine tools, robots, punches, presses, mills, and propulsion systems for electric vehicles are but a few examples of the application of electric machines in engineering.

Note that in Figure 14.1 we have explicitly shown the direction of two magnetic fields: that of the rotor $\mathbf{B}_R$ and that of the stator $\mathbf{B}_S$. Although these fields are generated by different means in different machines (e.g., permanent magnets, alternating currents, direct currents), the presence of these fields is what causes a rotating machine to turn and enables the generation of electric power. In particular, we see that in Figure 14.1 the north pole of the rotor field will seek to align itself with the south pole of the stator field. It is this magnetic attraction force that permits the generation of torque in an electric motor; conversely, a generator exploits the laws of electromagnetic induction to convert a changing magnetic field to an electric current. To simplify the discussion in later sections, we now introduce some basic concepts that apply to all rotating electric machines. Referring to Figure 14.2, we note that for all machines the force on a wire is given by the expression

$$\mathbf{f} = iw\mathbf{l} \times \mathbf{B}$$

Where iw is the current in the wire, $\mathbf{l}$ is a vector along the direction of the wire, and $\times$ denotes the cross product of two vectors. Then the torque for a multi-turn coil becomes

$$T = KBiw\sin \alpha$$

Where

B = magnetic flux density caused by stator field

K = constant depending on coil geometry

α = angle between $\mathbf{B}$ and normal to plane of coil

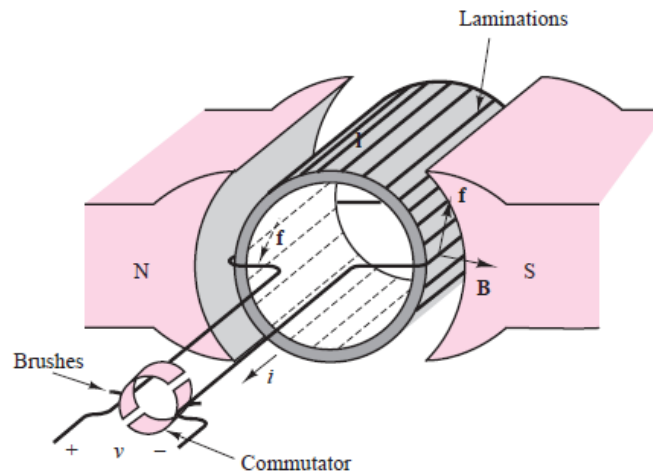


Figure 14.2 Stator and rotor fields and the force acting on a rotating machine

In the hypothetical machine of Figure 14.2, there are two magnetic fields: one generated within the stator, the other within the rotor windings. Either (but not both) of these fields could be generated by a current or by a permanent magnet. Thus, we could replace the permanent-magnet stator of Figure 14.2 with a suitably arranged winding to generate a stator field in the same direction. If the stator were made of a toroidal coil of radius R (see Chapter 13), then the magnetic field of the stator would generate a flux density B , where

$$B = \mu H = \frac{\mu N i}{2\pi R}$$

and where N is the number of turns and i is the coil current. The direction of the torque is always the direction determined by the rotor and stator fields as they seek to align to each other (i.e., counterclockwise in the diagram of Figure 14.1). It is important to note that Figure 14.2 is merely a general indication of the major features and characteristics of rotating machines. A variety of configurations exist, depending on whether each of the fields is generated by a current in a coil or by a

LECTURE-6

CONCEPTS DC MACHINE

FUNDAMENTAL:

Emf induced in coil $e = -N \frac{d\Phi}{dt}$ volt

Emf induced in a conductor $e = Blv$ volt

Force developed in a conductor $F = BIL N$

Electrical energy $E_n = VI t$ Whour

Electrical power (Active) $P = VI$ Watt

Resistance of conductor $R = \rho l/a$ Ohm

Eqn. of current by Ohm's law, $I = V/R$ A

Magnetizing force, $H = NI/L$ AT/m

Flux density, $B = \Phi/A$

Electrical field intensity, $E = V/d$ V/m

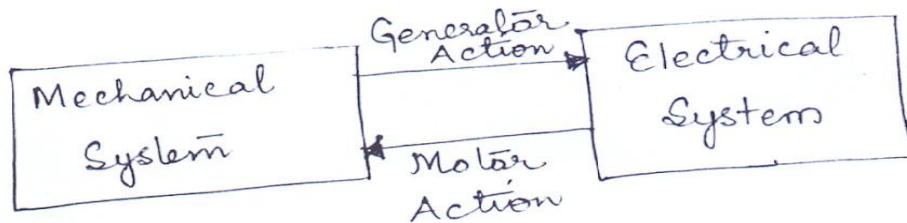
Current density, $J = I/a$ A/m²

INTRODUCTION :

DC machine is a device which converts mechanical energy into electrical energy and vice versa.

When the device acts as a generator (or dynamo), mechanical energy is converted into electrical energy.

On the other hand, when the device acts as a motor, electrical energy is converted into mechanical energy.



ESSENTIAL FEATURES:

There is no real difference between a dc. motor and dc generator either from the point of view of the essential components or of the fundamental principles involved.

In the generator there is a conversion of mechanical to electrical energy while in the motor there is the reverse conversion of electrical to mechanical energy. The structure difference is merely one of the enclosure. Most generators work in sheltered situations e.g. in power stations and an open type of construction is thus possible, the advantage being that all the parts are in consequence, easily accessible. Motors often work in exposed

Situations, where there are dirt as damp, chemical fumes, explosive gases and so on, and it may therefore be necessary to enclose all the working parts.

CONSTRUCTION:STATORROTOR

Yoke/Frame

Armature core

Field magnet
(pole core & pole shoe)

Armature winding/ conductor

Main field winding

Commutator & Brushes

Interpoles/commutating poles

Shaft

Interpoles winding

Bearings

STATOR & ROTOR

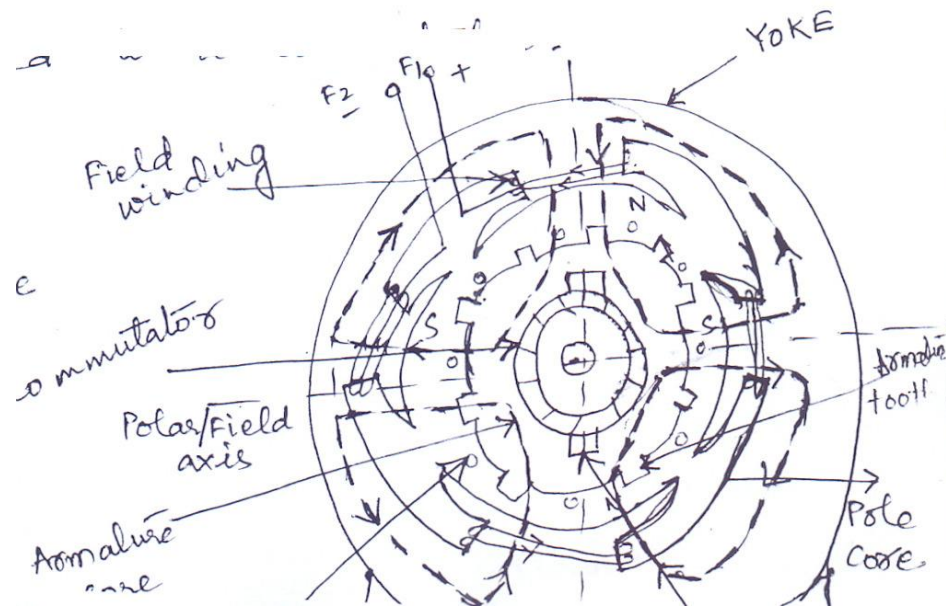
All conventional electrical machines consist of a stationary member called the stator separated by an air gap from a rotating member called rotor.

In d.c machines the stator usually consists of salient poles with coils wound round them so as to produce a magnetic field.

The rotor is familiarly called the armature and consists of a series of coils located in slots around its periphery and connected to a commutator.

Yoke/Frame

Yoke is the outer frame dc m/c. It carries the magnetic flux provided by the pole and acts as a protecting shield for the entire machine. In small generators yokes are made of cast iron whereas in large machine cast sheet is used.



Field magnet :

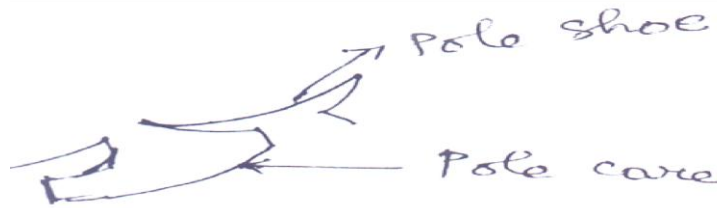
It is a strong permanent magnet (in case of a small dynamo) or an electromagnet (in case of large dynamo) of intense magnetic field.

Pole core & Pole shoe : The field magnet has two parts

Pole core (pole pieces)

Pole shoes.

Pole core is made of cast steel or cast iron with laminated pole shoes screwed on to the holes in the yoke.



Pole shoes spread out the flux in the air gap and reduce the reluctance of the magnetic path due to its large cross -section.

Poles shoes support the exciting coils.

Pole coils (field winding)

- It is otherwise known as exciting winding
- These are fine copper wire (or strip) wound around the pole pieces
- The flux produced by the winding is cut by the revolving armature

ARMATURE CORE:

- It houses armature coils in the slots. It is cylindrical or drum shaped.
- Armature is placed in between the two poles of field magnet and is rotated about its central axis mechanically (by a prime mover)



- It is usually made of circular sheet steel disc or laminations.

- Thickness of laminations is of the order of 0.5mm. Perforations exist in these laminations to provide axial flow of air through the armature for cooling purposes.
- Armature core is keyed to the shaft.
- During rotation it cuts the magnetic flux of the field magnets.
- It also provides a path of very low reluctance to the flux from north pole to south pole.

ARMATURE WINDING:

- It is made up of copper.
- It consists of large no. of insulated coils, each coil having one or more turns.
- Armature conductor are placed in armature slots.

Commutator:

- It converts the alternating current produced in the armature conductors into direct current.
- It consists of wedge shaped copper segments (Insulated from either side to form a ring. The number of segments of commutator equals the no. of armature coils.
- It facilitates the collection of current from the armature conductors.

Brushes

- They are fixed on the commutator by pressure springs.
- They are usually made of a high grade carbon or graphite and are in the shape of rectangular block.

- These are housed in a brush holder.
- They collect the current from the commutator and finally they pass on the EMF generated to load.

Bearing

- The armature shaft is supported at the commutator end on ball or roller bearings which are packed in hard oil.
- They provide quieter operation, together with reduced wear and tear.
- Ball bearings are frequently employed for their reliability.
- Roller bearings are preferable for heavy duties.

Interpoles

- In addition to the main poles, the dc machines are usually fitted with auxiliary poles placed between the main poles and they are called interpoles, the purpose of which is to provide better commutation.
- It minimizes sparking at the commutator.

Types of Armature Winding

These are of two types : Lap winding

Wave winding

Pole Pitch

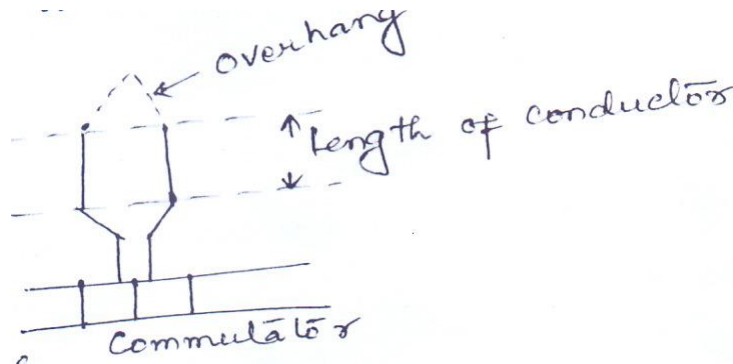
- It is defined as the no of armature conductors per pole.
- If there are 80 conductors and 8 poles, pole pitch is $80/8 = 10$.

Conductor

- It is defined as the length of a wire lying the magnetic field as shown in fig1.

Coil

- Two conductors with their end connections shown in fig1 from one coil.



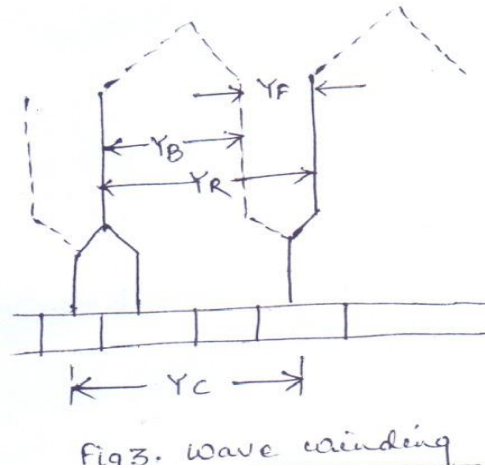
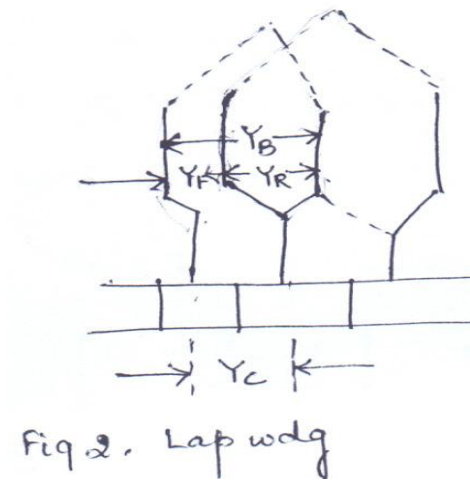
- Coil may be single turn coil or multi-turn coil.
- End connections are called overhang.

Coil Pitch

- It is defined as the distance between two sides of a coil in terms of armature slots.
- It is defined as the periphery of the armature between two sides of a coil.
- If the coil pitch is equal to the pole pitch, the winding is said to be full pitched, otherwise it is half-pitch

Pitch:

It is defined as the distance between the second conductor one coil and the first conductor of the next coil as shown in fig.2 & 3.



Back Pitch :

It is defined as the distance measured in terms of armature conductors that a coil advances on the back side of the armature as shown in fig.2 & 3.

Resultant Pitch: (Y_R)

It is defined as the distance between the beginning of one coil and the beginning of the next coil or

lap winding $Y_R = Y_B - Y_F$

wave winding $Y_R = Y_B + Y_F$

Coil Pitch (Y_C):

It is defined as the distance between the beginning of one coil to the beginning of the next coil as shown in fig 2 & 3 for lap and wave winding respectively.

Lap winding:

To get these type of following rules are commonly used: winding must be full pitched i.e.

$$Y_B + Y_F = \text{Pole Pitch}$$

1. To place the coil properly on the armatures, the front pitch as well as back pitch must be odd.
2. No. of commutator segment = no. of coils.
3. winding must close upon itself.
4. $Y_C = \pm 1$

For Progressive or right handed winding

$$Y_F = Z/P - 1 \quad \text{and} \quad Y_B = Z/P + 1$$

For Retrogressive or left handed winding

$$Y_F = Z/P + 1 \quad \text{and} \quad Y_B = Z/P - 1$$

Wave Winding:

1. Y_B & Y_F are odd and of opposite sign.
2. Y_B & Y_F are nearly equal to the pole pitch and differ by 2.
3. $Y_C = (\text{No. of commutator bars} \pm 1) / \text{No. of pairs of pole}$

Comparison:

Lap winding	Wave winding
No. of parallel paths = no. of poles	No of parallel path = 2
No. of brushes sets = no. of poles	No. of brushes sets = 2
Used for high current & low voltage	Used for low current & high voltage

PRINCIPLE OF OPERATION OF DC GENERATOR:

LECTURE-7 DC GENERATOR

Principle:

Faradays law of electromagnetic induction according to which emf is induced in the conductors which drives the current through the conductor.

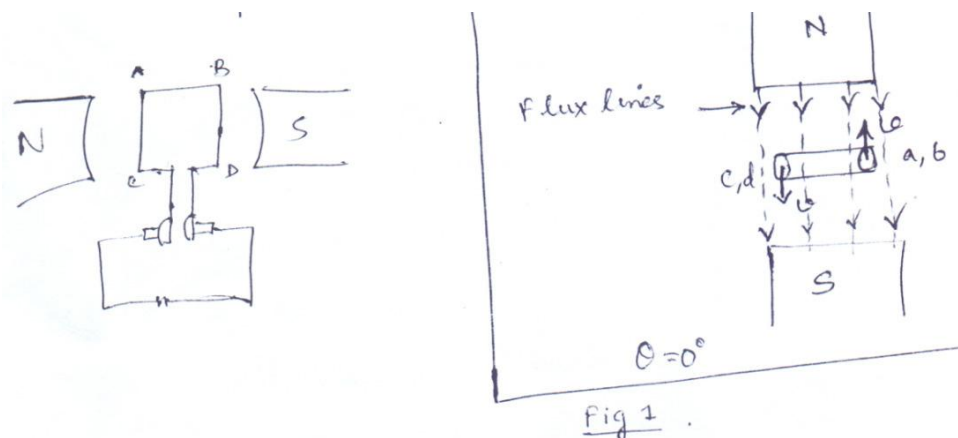
Fig below shows a single turn generator.

The coil is rotated in anticlockwise direction with the help of prime mover.

Fig. shows the different instants of the induced emf due to different positions of the coil.

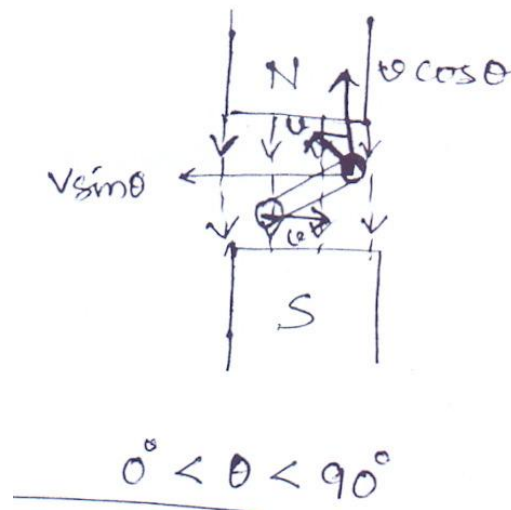
Magnetic field is produced either by a permanent magnet or an electromagnet energised by the DC supply.

When the angle $\theta = 0^\circ$, the coils is perpendicular to the magnetic field and the instantaneous component of velocity of conductors ab and cd is parallel to the magnetic field.

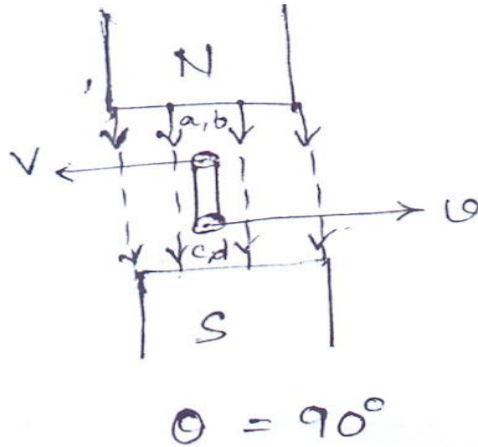


The flux linked with the coil is maximum but the rate of change of flux is minimum. Hence emf cannot be generated. Therefore current flows through the conductors.

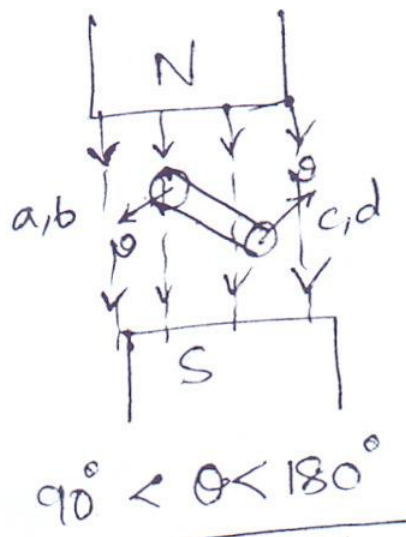
When $0 < \theta < 90^\circ$, due to the rotation of the coil in the anticlockwise direction, the velocity of the conductor has two components that is $v \sin \theta$ and $v \cos \theta$. Therefore the conductors will cut the flux due to $v \sin \theta$. Hence emf is induced in the conductors and current flows in the conductors.



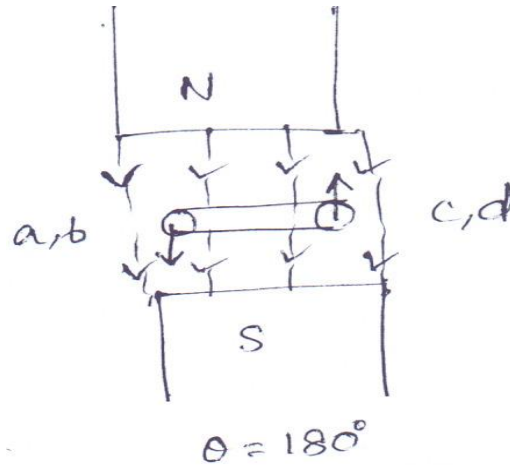
If when $\theta = 90^\circ$, coil is parallel to the magnetic field and the instantaneous component of velocity of conductors ab and cd is perpendicular to the magnetic field. The flux linked with the coil is minimum but the rate of change of flux is maximum. Hence emf generated is maximum. Therefore maximum current flows through the conductors.



When the coil rotates from $\theta = 90^\circ$ to $\theta = 180^\circ$, the components of velocity that is $\sin$ of conductors perpendicular to the field decreases and hence emf is induced in the conductor and current flow in the conductors.

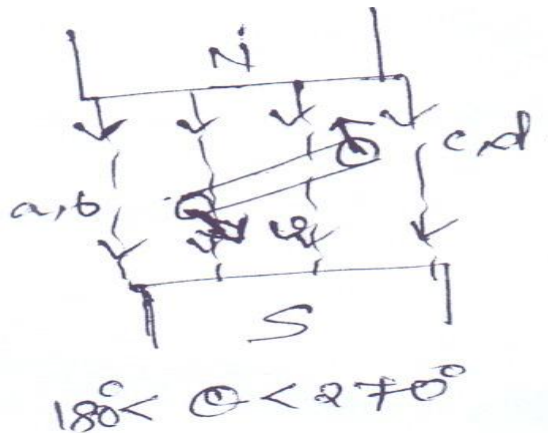


When $\theta = 180^\circ$, the coil is perpendicular to the magnetic field and the instantaneous components of velocity of conductors ab and cd is parallel to the magnetic field.

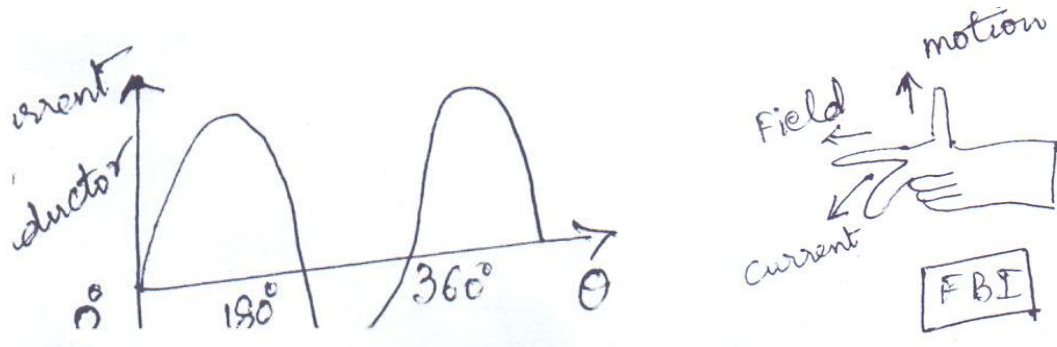


The flux linked with the coil is maximum but the rate of change of flux is minimum. Hence emf cannot be generated. Therefore the current flows through the conductor.

When $180^\circ < \theta < 270^\circ$ due to rotation of the coil in the anticlockwise direction, the velocity of the conductors has two components that is $v \sin$ and $v \cos$ and the direction is opposite to the case when $0^\circ < \theta < 90^\circ$. Therefore the conductor have the flux due to $v \sin$.

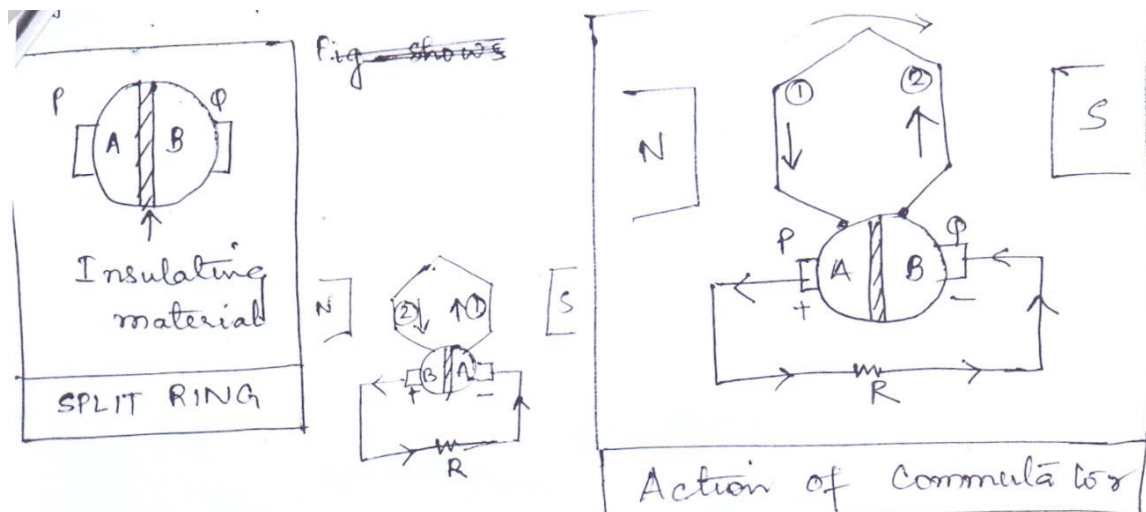


Hence emf is induced in the conductors in the direction opposite to the cases of $0^\circ < \theta < 90^\circ$ and current flows in the conductors in the direction opposite to the $0^\circ < \theta < 90^\circ$. Similarly the other case till $\theta = 360^\circ$.



The direction of flow of current through the conductor can be found by Fleming's right hand rule. To have a unidirectional current a commutator is used.

Fig below shows a split ring where two segments A and B are separated by an insulating material. Let the conductor 1 be connected to segment A and conductor 2 be connected to segment B.

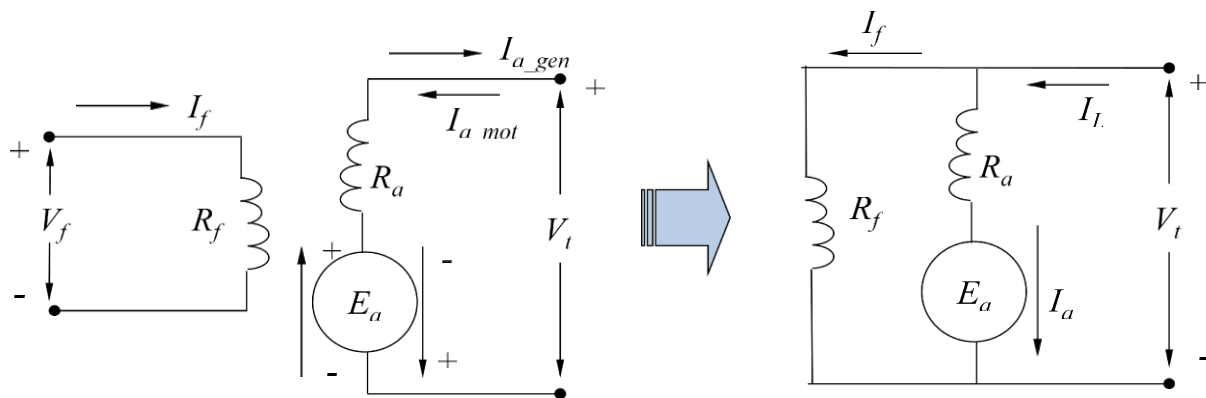


The conductor 1 carries current in the downward direction while the conductor 2 carries current in the upward direction and current through the resistance R flows

from P to Q. After one half of rotation, conductor 1 and 2 come under the influence of south and north pole respectively.

The split ring is mounted on the shaft and it rotates with the armature. The split ring will reverse this positions with the reversing of the position of the conductors, but the direction of current flow through the conductor remains unaltered because the P & Q are stationary.

Equivalent circuit of a DC Machine



$$V_f = I_f R_f$$

$$V_t = E_a \pm I_a R_a$$

Generated EMF and Electromagnetic Torque

$$V_f = I_f R_f$$

$$V_t = E_a \pm I_a R_a$$

LECTURE-8

TYPES OF DC MACHINES

Motor: $V_t > E_a$

Generator: $V_t < E_a$

Voltage generated in the armature circuit due the flux of the stator field current:

$$E_a = K_a \phi_d \omega_m$$

K_a : design constant

Electromagnetic torque:

$$T_e = K_a \phi_d I_a$$

$$P_{em} = E_a I_a = T_e \omega_m$$

Types of DC Machines

Both the armature and field circuits carry direct current in the case of a DC machine.

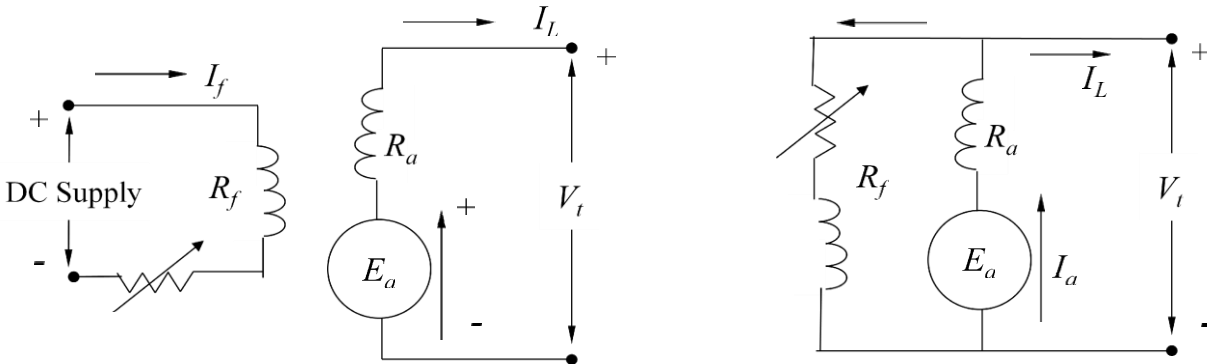
Self-excited DC machine: when a machine supplies its own excitation of the field windings. In this machine, residual magnetism must be present in the ferromagnetic circuit of the machine in order to start the self-excitation process.

Separately-excited DC machine: The field windings may be separately excited from an external DC source.

Shunt Machine: armature and field circuits are connected in parallel. Shunt generator can be separately-excited or self-excited.

Series Machine: armature and field circuits are connected in series.

Separately-Excited and Self-Excited DC Generators



Example: A 100-kW, 250-V DC shunt generator has an armature resistance of 0.05 Ω and field circuit resistance of 60 Ω . With the generator operating at rated voltage, determine the induced voltage at (a) full load, and (b) half-full load.

Ans. (a) At full load,

$$V_t = E_a - I_a R_a$$

$$I_f = 250/60 = 4.17 \text{ A}$$

$$I_{L_FL} = 100,000/250 = 400 \text{ A}$$

$$I_a = I_{L_FL} + I_f = 400 + 4.17 = 404.17 \text{ A}$$

$$E_a = V_t + I_a R_a = 250 + 404.17 \times 0.05 = 270.2 \text{ V}$$

(b) At half load,

$$I_f = 250/60 = 4.17 \text{ A}$$

$$I_{L_HL} = 50,000/250 = 200 \text{ A}$$

$$I_a = I_{L_HL} + I_f = 200 + 4.17 = 204.17 \text{ A}$$

$$E_a = V_t + I_a R_a = 250 + 204.17 \times 0.05 = 260.2 \text{ V}$$